

Seeing Is Deceiving

● Yellow Level: Ready for a Challenge

Name: _____ Date: _____ Period: _____

The Science of Seeing

Your eyes collect light, but your **brain** decides what you're seeing. Your brain uses shortcuts to process information quickly—but these shortcuts can be fooled. That's why optical illusions work!

Key Concept: Perception isn't passive—your brain actively constructs reality based on patterns, expectations, and context. Artists have used this for centuries to create depth, movement, and impossible images.

Activity 1: Analyze an Illusion

Examine the optical illusion shown in class. Sketch it below and analyze how it works:

Sketch the illusion

Analysis Questions:

What visual trick is being used? (contrast, perspective, pattern, context, etc.)

What assumption is your brain making that causes the illusion?

How might an artist use this technique intentionally?

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Activity 2: Design Your Own Optical Illusion

Create an original optical illusion using one of these techniques:

- **Ambiguous figures:** Images that can be seen as two different things
- **Size/perspective illusions:** Objects that appear different sizes due to context
- **Motion illusions:** Static patterns that appear to move
- **Impossible objects:** Shapes that couldn't exist in 3D space

Design your optical illusion here

Explain your design:

What technique did you use?

What assumption does your illusion exploit?

Mini Museum Connection

The Victorian era (1837-1901) saw an explosion of optical toys and illusions. Visit minimuseumproject.com/exhibitions/seeing-is-deceiving/illusions to see how 19th-century inventors used these same principles!